

Taylor Swift Lyrics from 2006 to 2020 - Shelton Wright & Stefani Morgan

Introduction

Taylor Swift, according to New York Times Magazine, is one of the “30 Greatest Living American Songwriters” today (How Taylor Swift Writes a Song). In an interview she stated her famed lyrics are inspired by stories, love, and the warnings and lessons of her life (How Taylor Swift Writes a Song).

Our goal with this project was to look at how the emotional themes of love, friendship, and heartbreak show up across the songs she has written. We hoped to identify how they changed over the course of her albums, and therefore over time.

Data

- Sourced from a curated dataset from the Tidy Tuesday Project located in the GitHub repository `textmining-infopros`.

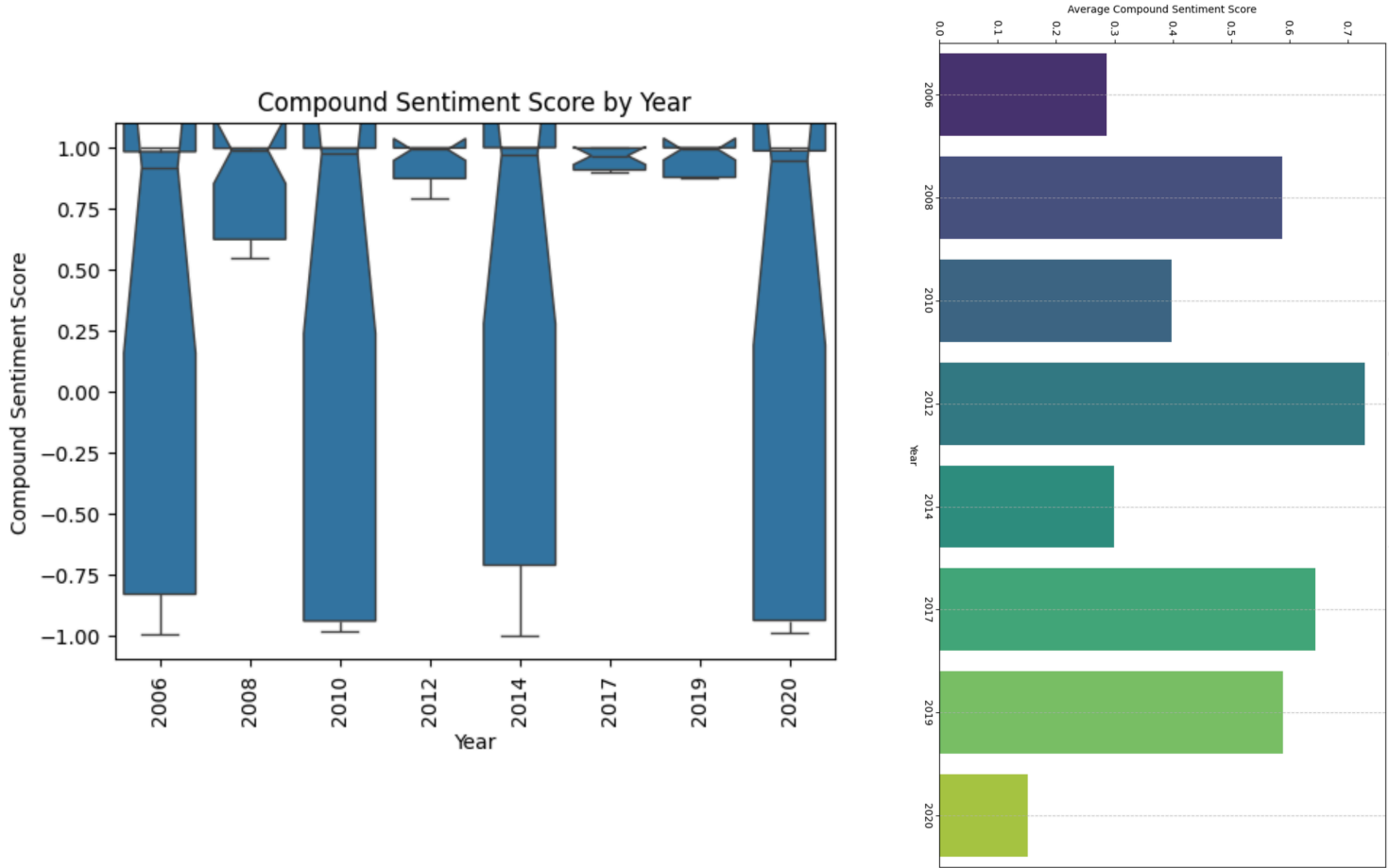
Method

- Downloaded the cvs file from the TidyTuesday GitHub repo
- Preprocessed the text by removing punctuation, making all characters lowercase, and removing stopwords
- Performed topic modeling with LDA models to analyze the main topics/clusters of words in the lyrics
- Performed sentiment analysis with VADER to analyze the average emotions of each album

References

- https://github.com/rfordatascience/tidytuesday/blob/main/data/2020/2020-09-29/taylor_swift_lyrics.csv#L32
- “How Taylor Swift Writes a Song, In Her Own Words.” The New York Times Magazine, 28 Apr. 2026, <https://www.nytimes.com/2026/04/28/magazine/taylor-swift-songwriting-process-interview.html>.

GitHub Repository https://github.com/Sigurd-RW/lis4693/blob/main/final%20project/Final_Project.ipynb



Findings

From the sentiment analysis we found a trend that Taylor Swift’s lyrics seemed to alternate between being more negative and more positive. We were able to calculate sentiment scores for each album and correlate that to the years they were released to show the change over time. Then with topic modeling we generated 10 main topics to identify the kinds of emotional themes that are present in Taylor Swift’s lyrics. The 4 topics on display show that they can vary from heartbreak, to regret, to growth and the future, and to religion respectively. These insights about the topics she writes about offer some context to the sentiment scores generated for each album.

Conclusion

Topic modeling and sentiment analysis allowed us to analyze the lyrics of all of Taylor Swift’s songs to gain insights about her use of emotional themes over time. Our analysis revealed that the meanings of her songs can range from both negative and positive extremes, but the trend was that albums typically alternated on that. During our case study we did come across some limitations in the dataset that might have influenced the accuracy of our results. While we were able to overcome the absence of data on the years of each album from the dataset, we struggled determining which words to filter our during the text processing to generate accurate topics. We removed many ad-libs and overly repeated words, but weren’t sure on words that were frequent in every song but seemed to pertain to the topics. Changing which custom stop words you include would influence the outcome of this case study however.

